

GETACHEW AMBAYE, Ph.D.

Robotics & Automation Engineer — Manufacturing Systems — AI-Driven Engineering

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🌐 linkedin.com/in/getachew-ambaye 📁 Portfolio

Professional Summary

Engineering professional with expertise in robotics, automation, advanced manufacturing, machine vision, finite element analysis, and AI-driven engineering systems. Experienced in PLC programming, CAD/CAM, additive manufacturing, experimental system design, and interdisciplinary research. Strong background in smart manufacturing, intelligent sensing, and hands-on engineering problem solving.

Technical Skills

CAD/CAM: SOLIDWORKS, CATIA V5, 3DEXPERIENCE, Fusion 360, AutoCAD, Inventor, GD&T

Simulation: ABAQUS, ANSYS, MSC ADAMS, SIMULIA, FEA, Dynamic Analysis

Automation: Siemens TIA PLC, CLICK PLC, Sensors, Actuators, ABB Robots, UR Robots

Programming: Python, MATLAB, OpenCV, C, Structured Text

Manufacturing: CNC, Machining, Metrology, Welding, 3D Printing, Hydraulics, Pneumatics

Professional Experience

Assistant Professor – Austin Peay State University 2025–Present

- Teach manufacturing, automation, CAD/CAM, robotics, and mechatronics related courses.
- Developed laboratory systems involving PLCs, machine vision, robotics, sensors, and additive manufacturing.
- Lead applied engineering projects integrating AI, VR, and smart manufacturing technologies.

Graduate Research Assistant – Wichita State University 2022–2025

- Conducted research in soft robotics, AI, machine vision, and advanced manufacturing.
- Developed experimental systems for vibration analysis, actuator characterization, and intelligent sensing.
- Applied deep learning and computer vision methods for manufacturing defect detection and inspection.

Engineering Graphics Instructor – Wichita State University 2022–2025

- Taught CAD modeling, engineering graphics, and product design using CATIA and 3DEXPERIENCE.

Selected Projects

- Developed AI-based soft robot workspace prediction using FEA and neural networks.
- Designed machine vision systems for drilled-hole misalignment and screw identification.
- Improved 3-axis milling machine stiffness using vibration and finite element analysis.
- Designed sensor-integrated CAM analyzer system with embedded programming.
- Developed industrial pipe-pulling systems using dynamic and structural analysis.

Education

Ph.D., Industrial Systems and Manufacturing Engineering Wichita State University, 2025

M.Sc., Mechanical Design Engineering Jimma Institute of Technology

B.Sc., Mechanical Engineering Jimma Institute of Technology

Selected Publications

- Soft Robot Workspace Estimation via Finite Element Analysis and Machine Learning
- Detecting Misalignments of Drilled Holes using Machine Vision
- Robot Link Damage Detection using Deep Learning and Vibration Analysis
- Soft Robot Design, Manufacturing, and Operation Challenges: A Review

Awards

Outstanding Graduate Student Award (2025) Best Paper Award – ICAST 2023